

Leveraging machine learning methods to predict COVID-19 vulnerability in U.S. counties based on socioeconomic factors

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ABSTRACT: As COVID-19 gained pandemic status, the number of confirmed cases in the US surpassed that of all other countries. Although the virus spread throughout the US, not all areas were affected equally. This retrospective study aims to explore these inequalities through pre-pandemic socioeconomic characteristics by attempting to create a predictive model for COVID-19 vulnerability at the county level. A total of 103 features of socioeconomic data for 2610 US counties (out of a total of 3007) were sourced from various online databases such as the US Census Bureau, the US Department of Agriculture, and the Association of American Medical Colleges. Additionally, to quantify each county's COVID-19 vulnerability, we defined 3 custom measures: incidence, mortality, and case fatality. These measurements were calculated using case and death data taken 29 days after each county's first case. Machine learning classification algorithms – including random forest, multi-layer perceptron neural network and XGBoost – were then used to predict the incidence, mortality, and case fatality of US counties. Through analysis, we were able to predict a county's COVID-19 incidence with ~47% accuracy, mortality with ~59% accuracy, and case fatality with ~61% accuracy by looking solely at pre-pandemic socioeconomic factors. A list of important features was extracted using a built-in XGBoost function for each vulnerability measure (incidence, mortality, and case fatality). Many of these features are typically associated with pandemic spread (e.g., population density and medical infrastructure), while other features were unexpected (e.g., education) and warrant further studies to identify their role in disease propagation. Furthermore, the difficulties our model experienced support the notion that region-specific policies play an important role in successfully mitigating this crisis. The moderate success achieved in this study proves the feasibility of using classifiers as a pandemic preparedness evaluation tool.

KEYWORDS: COVID-19, COVID-19 vulnerability, machine learning, predictive model

INTRODUCTION

COVID-19, a novel disease caused by SARS-CoV-2, first appeared in Wuhan, China on December 31, 2019. As of October 18, 2020, there have been 40,117,591 confirmed cases and 1,114,736 deaths worldwide [1]. The United States (US) alone has experienced at least 8,387,047 confirmed cases and 222,447 deaths [2]. With over 216 countries, areas, and

territories afflicted, the international impact of COVID-19 cannot be understated. Due to the rapid infection rate of COVID-19, a firm understanding of its spread is essential to better equip communities in keeping their citizens healthy.

Numerous groups have reported varying success in developing models to characterize the COVID-19 pandemic. These included studies to identify risk factors for COVID-19 [3-4], project the trajectory of cases [5-6], and predict individuals

and counties most vulnerable to COVID-19 in the US [7].

Using COVID-19 data from March 14 through March 31, 2020, Mehta et al. reported a sensitivity of 71.5% and specificity of 94% when predicting the likelihood of US counties going from zero to at least one documented case within a 5-day period. An XGBoost classifier model and an XGBoost regression model were used to predict county vulnerability [3]. Likewise, Hamzah et al. described a tool to analyze public sentiment to predict COVID-19 trends. This tool utilized Python's `scipy.integrate.odeint` (ordinary differential equation integration) library to plot trajectories for exposed and infected individuals [4]. Another popular avenue for analysis has been forecasting COVID-19 case trends. Some studies have predicted metrics such as when national COVID-19 outbreaks could be expected to end [5-6]. Biswas et al. applied this approach to data from China using a susceptible, infectious, and/or recovered (SIR) model, while Petropoulos et al. considered the global situation using a time series forecasting approach [5-6]. Lastly, some research have aimed to identify populations that are most vulnerable to COVID-19 [7,23]. In particular, DeCaprio et al. proposed 3 unique models to help identify individuals most susceptible to COVID-19, making use of logistic regression models and gradient-boosted trees [7]. Successful county response to public health emergencies relies on a combination of strong existing infrastructure and appropriate steps taken as a reaction to the emergency. This study aims to use classification algorithms to construct a model that can predict a county's COVID-19 vulnerability based on pre-pandemic socioeconomic data. Our study builds upon the methodologies used in the papers mentioned above. First, by incorporating a broader range of characteristics such as economic indicators, crime statistics, demographic information, and population health, we can consider factors that may not commonly be hypothesized to affect COVID-19 spread. Second, by considering a customized timeframe for each county, we can achieve a more accurate depiction of the dynamic situation. Third, rather than projecting COVID-19 case statistics forward in time using modeling techniques, this retrospective study aims to explore the pre-pandemic socioeconomic characteristics of US counties to create a locally predictive model for COVID-19 vulnerability. Additionally, this study considers a variety of statistical learning algorithms, such as k-nearest neighbours, random forest, multi-layer perceptron neural network and XGBoost, to determine the

model with highest efficacy at identifying at-risk counties. A model of this nature would be valuable to health authorities to plan and prepare for possible recurrences of COVID-19 and even future pandemics with similar patterns. Lastly, by focusing on county level data as opposed to state level data, this study provides a more holistic account for characteristic nuances between smaller regions. Therefore, in the case that socioeconomic differences between counties are related to COVID-19 incidence levels, at-risk communities can be more accurately predicted. Through the results of this study, both state and health authorities can move towards implementing preventative policies to improve their existing infrastructure.

METHODS

All analysis scripts, data processing scripts, and associated source data referenced in the following materials and methods section can be found in the GitHub repository (<https://github.com/clo514/stem-fellowship-bdc-2020>).

Sourcing the dataset

The county-level socioeconomic data used in this study was publicly sourced from the databases of the following organizations: the US Census Bureau [8-10], the U.S. Department of Commerce [11-12], the US Department of Agriculture [13-15], the Kaiser Family Foundation [16], the Association of American Medical Colleges [17], the New York Times [18], as well as independent contributions [19-20].

From these databases, a total of 103 data fields for 2610 US counties (out of 3007 total counties) were extracted to be used as features for classification. A small number of counties had insufficient data to fulfill all 103 data fields and were dropped from analysis. A comprehensive list of each feature and its description can be found in the appendix. These 103 features include: quantitative data on economic indicators, including gross domestic product and median income; medical personnel and infrastructure, including number of active physicians and number of intensive care unit (ICU) beds; population demographics, including population density and age demographics; population health statistics, including obesity rate and alcoholism rate; and crime statistics, including number of murders and number of arson-related crimes.

Time series data for COVID-19 at the county-level was sourced from the New York Times [18]. This dataset was based

on reports from state and local health agencies. It reports the cumulative number of infections and deaths for each county from January 21 to June 27, 2020.

Quantifying vulnerability

In the absence of a standardized metric of vulnerability, we describe three measures: incidence, mortality, and case fatality. These measures are modified versions of the formal epidemiological definitions of the same name. All later mentions of incidence, mortality and case fatality in this paper will refer to the below definitions specifically. For each county, these measures are defined by Equations 1-3. Note that the total cases and deaths indicate the cumulative case and death counts 29 days after the first confirmed case in each county.

Incidence, mortality, and case fatality are common measures of vulnerability in epidemiological studies. Traditionally, these 3 measures follow the same formulae as those previously described. These formulae traditionally use total case and death counts of a given geographic location. However, we recognize that each county is at a different stage of their COVID-19 response, with some counties receiving their first positive case much earlier than others. It may therefore be inaccurate to compare each county's COVID-19 vulnerability purely based on an arbitrary day's cumulative data. To standardize all the counties, case and death counts were examined 29 days after each county's first case. Thus, the three measures are based on how infection spread after a county's initial outbreaks, rather than a county's status at a certain point in time.

A logarithmic scale was used in response to the skewness introduced by large denominators compared to numerators. The rounding operation was used to ensure that each property is integer-valued so classification can be performed. The incidence scale ranges on $[-10, 0]$, the mortality scale ranges on $[-14, 0]$, and the case fatality scale ranges on $[-7, 0]$. In each of these scales, lower values indicate lower levels of each measure, while higher values indicate greater levels of each measure. Values less than the lower bound of each scale were rounded to the lower bound. For example, a value less than -14 on the incidence scale is rounded to -14, and this value indicates a low number of cases per population on day 30.

$$\text{Incidence} = \text{round}\left(\ln \frac{\text{total cases}}{\text{total population}}\right) \quad (1)$$

$$\text{Mortality} = \text{round}\left(\ln \frac{\text{total deaths}}{\text{total population}}\right) \quad (2)$$

$$\text{Case fatality} = \text{round}\left(\ln \frac{\text{total deaths}}{\text{total cases}}\right) \quad (3)$$

Data preprocessing

The data fields mentioned above were compiled into a single comma-separated value (CSV) file and imported into Jupyter Notebook for analysis. All 103 feature columns were normalized (demeaned and scaled by standardized deviation) prior to classification. The data was then split into two sets 90% ($n = 2349$) of the data points were used for training and validation, and the other 10% ($n = 261$) were used as a test set. After compiling our data, we consolidated it into a single, manageable `pandas.DataFrame` object compatible with the various classifiers. Furthermore, as some classifiers benefit from a processed dataset, the original dataset was normalized using `numpy.mean()` and `numpy.std()` functions.

Model selection and evaluation

Multiple classifier models were created for each vulnerability scale (incidence, mortality, and case fatality) using classifiers from the `sklearn` library and `xgboost` package [21-22]. Model selection was performed using threefold cross-validation. The choice of 3 splits is due to the least populated class only having 3 members. Select hyperparameters of each classifier were adjusted and the best performing classifier was tested on the test set to assess model accuracy. See Appendix B for a list of all classifiers used and hyperparameters tweaked.

For result visualization of the best classifier on each vulnerability scale, the dataset was repartitioned into 2 equally sized shuffled subsets ($n = 1305$). The best classifier was then retrained on the training set and tested on the testing set to produce a confusion matrix and two-colored scatterplots: one plot colored by the true category of each county and the other plot colored by the predicted category of each county.

Assessing feature importance

Feature importance for each vulnerability scale (incidence, mortality and case fatality) was assessed using the `xgboost.get_fscore()` function. The score indicates how many times

a feature was used in deciding the final category of an observation.

RESULTS

Visualizing county vulnerability

Figure 1 shows the distribution of county scores for each level of incidence, mortality, and case fatality. Note that while the defined incidence scale is $[-10, 0]$, there are no counties with an incidence greater than -3 . For mortality, where the scale ranges on $[-14, 0]$, there are no counties with a mortality greater than -7 . Similarly, for case fatality, possible values range between $[7, 0]$, but there are no counties with a fatality greater than -1 .

Summary of results

For each vulnerability measure (incidence, mortality, and case fatality), we reported our model's predictive capabilities using the top 3 performing classifiers. All parameters associated with each reported classifier are included as well.

Predicting incidence

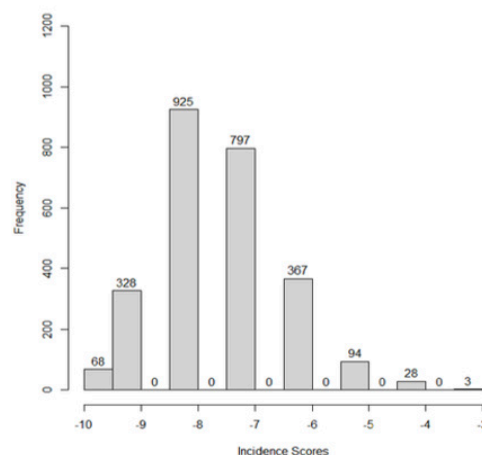
COVID-19 incidence was predicted most successfully using a random forest classification algorithm. The average cross validation score was 41.9% and the test score was 47.9%. XGBoost classification showed a 41.3% average cross validation score and 46.7% test score. A confusion matrix and scatter plot of counties labeled by true and predicted incidence (Figures 2-3) were also produced.

Predicting mortality

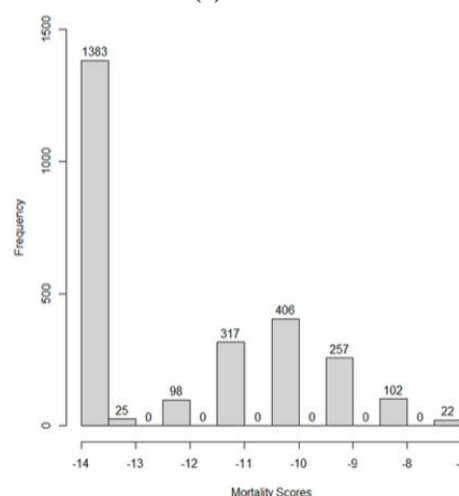
COVID-19 mortality was predicted most successfully using an XGBoost classification. The average cross validation score was 56.3% and the test score was 59.4%. Random forest classification algorithm showed a 57.7% average cross validation score and 59.0% test score. A confusion matrix and scatter plot of counties labeled by true and predicted incidence (Figure 3) were also produced.

Predicting case fatality

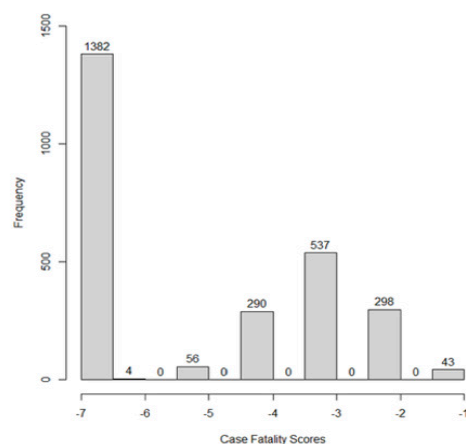
COVID-19 case fatality was predicted most successfully using an XGBoost classification algorithm, with an average cross validation score of 60.4% and a test score of 61.3%. Random forest classification showed an average cross validation score of 59.7% and a test score of 59.0%. A confusion



(a) Incidence



(b) Mortality



(c) Case Fatality

Figure 1. Frequency distribution of each vulnerability score: incidence, mortality, case fatality ($n = 2610$).

matrix (Figure 2c) and scatter plot of counties labeled by true and predicted incidence (Figure 3c) were also produced.

Table 1. Social factors of greatest positive correlation with phosphate concentration. The factor-region pairs that produced the greatest positive correlation coefficient with phosphate concentration.

Target	Classifier	Hyperparameters	Average Cross Validation Score	Test Score
Incidence	Random forest	$max_depth = 7$	0.419	0.479
	XGBoost	$max_depth = 3, min_child_weight = 2, n_estimators = 200$	0.413	0.467
	Quadratic discrimination analysis	uniform prior	0.473	0.452
Mortality	XGBoost	$max_depth = 1, min_child_weight = 4, n_estimators = 70$	0.563	0.594
	Random forest	$max_depth = 6$	0.577	0.590
	Neural network	identity activation function, stochastic gradient descent (SGD) solver, 100 hidden layers	0.560	0.590
Case fatality	XGBoost	$max_depth = 3, min_child_weight = 7, n_estimators = 50$	0.604	0.613
	Random forest	$max_depth = 9$	0.597	0.590
	Neural network	identity activation function, Adam solver, 20 hidden layers	0.580	0.590

Feature importance

The 15 most important features extracted from the xgboost.get_fscore() function for each vulnerability scale (incidence, mortality and case fatality) can be found in Table 2, in order from most to least important. The complete list of features ranked by importance for each vulnerability measure (i.e., incidence, mortality, and case fatality) can be found in Appendix B.

DISCUSSION

Interpretation of prediction scores

Given that the best performing classifier was XGBoost on case fatality with a 61.3% test accuracy score, we present two conclusions. First, case fatality can be partially predicted based on various socioeconomic factors. This is a reasonable conclusion as a county's ability to provide care for its sick directly influences the outcome of many cases. However, the shortcomings of this predictive rate may speak towards the underlying role of additional, non-socioeconomic factors, such as local, state or country policies instigated in response to the pandemic. Similar results can be seen with mortality and incidence. From the 59.4% and 47.9% respective test prediction rates, it can be concluded that a county's ability

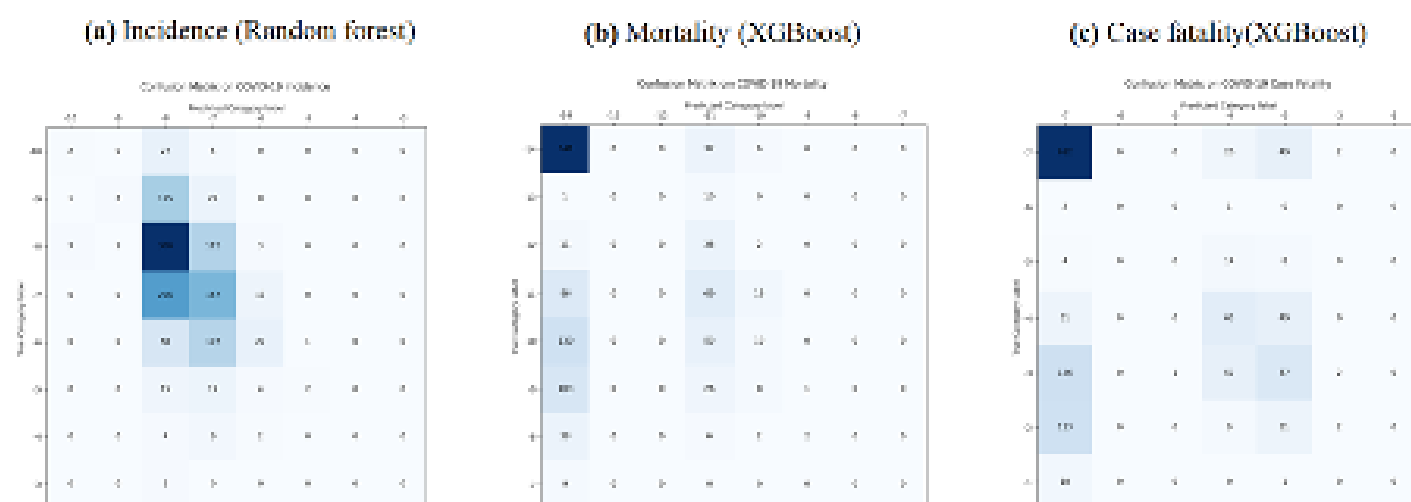


Figure 2. Confusion matrices for the best performing classifiers of each vulnerability measure (n = 1305).

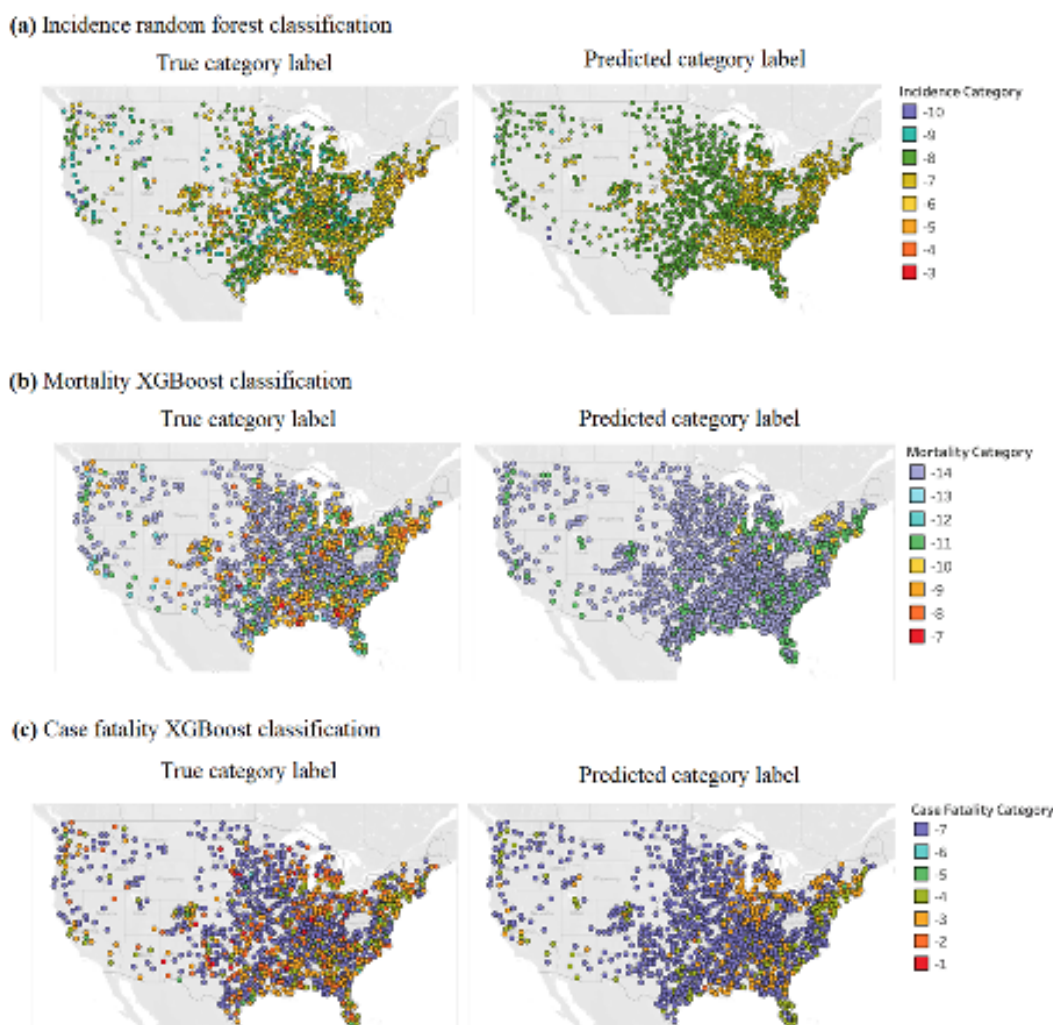


Figure 3. Scatterplots of counties colored by true and predicted target values, plotted based on geographical location ($n = 1305$). Mislabeling is represented by a difference in color of the same point across the two scatterplots.

to minimize the spread of disease seems to be partially related to socioeconomic factors. Again, however, the low rate highlights the potential importance of other factors not included in our study, such as government response, public sentiment, and policy.

Additionally, our results demonstrate the importance of documenting the symptomatic nuances of COVID-19. Failing to do so may drastically influence the outcome of analysis, especially when comparing counties with substantially different testing policies. The effects of such a discrepancy can be best observed by comparing each dataset's predictive power. While the aim of this study was to identify a single means of predicting COVID-19 vulnerability, it should be noted that overall, certain target datasets were more accurately predicted than others. From Table 1, both case fatality and mortality outperform incidence. In counties without

accessible and exhaustive testing, asymptomatic carriers of COVID-19 are less likely to be accounted for; symptomatic residents presumably have more reason to seek testing than both asymptomatic and healthy residents. Consequently, it is likely that the number of deaths attributed to COVID-19 is more accurately reported than the number of confirmed cases. Under this assumption, our classifiers are better trained to model mortality and case fatality, whose scales are based on the cumulative deaths in each county.

Considering the above, a top accuracy score of 61.3% shows promising feasibility in utilizing classifiers to identify at-risk populations during constantly evolving socioeconomic conditions. Public health authorities could build a county vulnerability model based on the current COVID-19 pandemic and employ it to predict an estimated county vulnerability in a future pandemic when socioeconomic characteristics have

Table 2. Top 15 most important features for the prediction of each vulnerability measure, as determined by `xgboost.get_fscore()`.

Rank	Incidence	Mortality	Case Fatality
1	Longitude	Population Estimate, 2019	Longitude
2	Latitude	Longitude	Latitude
3	Percent Excessive Drinking	Natural Population Change, 2019	Percent Smokers
4	Percent of Adults Completing Some College or Associate Degree, 2014-2018	Latitude	Percent Adults with Obesity
5	Income	Estimate of Medium Household Income, 2018	Otolaryngology Association of American Medical Colleges
6	Percent of Adults with Obesity	Percent of Adults Completing Some College or Associate Degree, 2014-2018	Estimate of Medium Household Income, 2018
7	Percent Physically Inactive	Active Primary Care Physicians per 10000 Population, 2018 AAMC	International Migration, 2019
8	Percent of Adults with less than a High School Diploma, 2014-2018	Larceny	Natural Population Change, 2019
9	Estimate of Medium Household Income, 2019	Otolaryngology AAMC	Percent Excessive Drinking
10	Percent Smokers	Economic topology, 2015	Percent of Adults with a High School Diploma only, 2014-2018
11	Natural Increase, 2019	Domestic Migration, 2019	Percent Physically Inactive
12	Natural Population Change, 2019	Obstetrics, Gynecology, Primary Care, 2019	Plastic Surgery AAMC
13	Percent of Adults with a Bachelor's Degree or Higher, 2014-2018	Female Ages 65+	Natural Increase, 2019
14	Percent of Adults with a High School Diploma Only, 2014-2018	Fraction of Active Physicians Who are International	Percent of Adults with less than a High School Diploma Only, 2014-2018
15	Percent Vaccinated	Quarterly Payroll	Percent of Adults Completing Some College or Associate Degree, 2014-2018

changed.

Interpretation of feature importance

The consistent appearance of several well-accepted pandemic contributing factors (e.g., population density, medical infrastructure, pre-existing health conditions) in the list of top 15 important features attests to the validity of the feature importance measure. This brings attention to the features that are not commonly considered to be pandemic predictors (e.g., educational attainment, natural population change). For example, the informative value of features related to population dynamics – including “Natural Increase 2019”, “Natural Population Change 2019”, and “Domestic Migration 2019” – might not be immediately evident. When we consider them as indicators of areas with shifting populations, the connection is clearer. Growing populations

are generally indicative of attractive communities to which individuals relocate to in search of job or other prospects. In turn, these areas become social hubs inherently more prone to outbreaks (as a product of a larger populous). Similarly, shrinking communities could potentially be less prone to case outbreaks. The impact of these newly discovered, hidden features on pandemic progression should not be overlooked in future studies.

A few unexpected features that repeatedly ranked in the top 15 were those related to education (e.g., “Percent of adults with a bachelor's degree or higher”, “Percent of adults with less than a high school diploma”, “Percent of adults completing some college or associate degree”). Thus far, education has been seldom investigated as a contributing factor of pandemic spread. Since positive and negative correlation cannot be determined from the feature importance list,

further studies should be undertaken to investigate the relationship between education attainment and disease propagation. One hypothesis could relate education to factors such as occupation, household size and income. The traits associated with these factors, such as precarious work environments, crowded housing conditions, limited remote alternatives and access to effective healthcare could lead to higher risk of infection. This is further supported by the presence of income-related fields in our list of top ranked features.

Moreover, it is interesting that positional features – primarily geographical latitude, and longitude – were consistently reported as important. One basic rationale is related to the proximity of communities and the transmission of an infectious disease. Indeed, a county's infected citizens have a greater chance of interacting with, and thereby infecting, their neighbours. As a result, aggregate county regions are more likely to share similar patterns of disease propagation. In a different vein, although it was not a focus of this study, latitude and longitude are related to climate, which could be another hidden feature worthy of investigation.

Improvements

The success of statistical learning methods relies largely on the accuracy, quality, and resolution of the dataset it is trained on. In future iterations of this study, it will be important to collect holistic and comprehensive data that accounts for socioeconomic characteristics potentially more relevant to the spread of COVID-19. For example, data pertaining to recent local air travel patterns or frequency of super-spreader events (e.g., parties, concerts, rallies) may elicit better predictive qualities for COVID-19 spread. Collecting relevant and complete data on this scale, however, is often a limitation beyond the control of data scientists.

Only small adjustments in a few hyperparameters were tested for each classifier to optimize performance. With more computational power and fine tuning, each classifier's parameters could be refined to strike a balance between overfitting and underfitting. Optimization of these classifiers in future iterations of this study have the potential to yield much higher predictive accuracies. In future iterations of this study, the research question could be reformulated as a regression problem to address the discretized binning of vulnerability numbers.

While the three vulnerability measures in this study (incidence, mortality, and case fatality) were able to show some efficacy when used with the statistical learning methods, they were ultimately crude representations of COVID-19 vulnerability. To increase the predictive accuracy of the statistical learning methods, a more nuanced and refined measure of vulnerability should be adopted. The true pervasiveness of COVID-19 may very well go beyond pure metrics (e.g., total cases, deaths, active). For example, pervasiveness may involve additional factors such as the acceleration of infection rates.

The lower predictive capabilities of these classifiers may be attributed to a poor spread of data across the different vulnerability categories. For example, due to the skewed distribution of COVID-19 incidence across the US, few counties have scores that fall at the extremes of the incidence scale. These score distributions lead to classifier bias towards the more populous category, as seen in the confusion matrices for incidence (Figure 1a), favoring the median values and neglecting the extreme values in their respective scales.

The same bias can be seen in the confusion matrices for mortality and case fatality (Figure 1), where the right-skewed distribution of both mortality and case fatality data led to classifier predictions favoring the lower values while neglecting the upper values. In future iterations of this study, a uniformly distributed scale should be used to train classifiers to attempt more accurate, less biased predictions.

A continuation of this study may consider using an abridged list of important features for a more classifier focused investigation for identifying at-risk communities. The important features from this study can serve as a foundation for future regression studies, which could explore the degree to which these features are associated with COVID-19 vulnerability. Further studies in this direction will aid in the creation of a robust predictive tool for guiding public health decision making.

CONCLUSION

As lockdown policies lift, apprehension surrounding a second wave of COVID-19 rises, prompting a need for studies such as this one. Identification of at-risk communities and verification of vulnerabilities in infrastructure are crucial for preparation and mitigation. Beyond the investigation of pre-pandemic characteristics, this study also demonstrated

the importance of dynamic factors, such as policy and rhetoric driven by a state's response to COVID-19. This study has established the groundwork for a potentially powerful predictive tool for public health authorities who are at the forefront of conceiving critical health policies. After implementing the improvements outlined in the discussion section, this model could become a reliable and regular part of community vulnerability analysis to improve pandemic preparedness. This model could even prove useful for the public, whose understanding of community vulnerabilities is also crucial to the reduction of pandemic spread. In addition, this study has identified critical, previously overlooked socioeconomic factors when it comes to pandemic modeling. Our findings suggest that future pandemic-related public health studies could consider incorporating education and human migration to the investigation process. Ultimately, this study seeks greater understanding of COVID-19 spread, and both the model's predictive success and shortcomings achieve this. In its success, we have demonstrated a proof-of-concept for predicting a county's susceptibility to COVID-19 through machine learning socioeconomic factors in a community. In its shortcomings, we have learned that a county's COVID-19 vulnerability may ultimately be at the mercy of the state and public's actions at the onset of spread.

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APPENDIX A

Table A1. Features and their associated descriptions.

Rank	Incidence	Case Fatality
0	Lat	Latitude
1	Long	Longitude
2	Population Estimate	Resident Total on 7/1/2019
3	NPOPCHG_2019	Change in resident total population 7/1/2018 to 2019
4	BIRTHS2019	Births in period 7/1/2018 to 6/30/2019
5	DEATHS2019	Deaths in period 7/1/2018 to 6/30/2019
6	NATURALINC2019	Natural increase in period 7/1/2018 to 6/30/2019
7	INTERNATIONALMIG2019	Net international migration in period 7/1/2018 to 6/30/2019
8	DOMESTICMIG2019	Net domestic migration in period 7/1/2018 to 6/30/2019
9	NETMIG2019	INTERNATIONALMIG2019 + DOMESTICMIG2019
10	CommutingIn	Number of workers commuting into a county for work (2017)
11	TotEmp	Total Mid-March employees with noise (2017)
12	QPayroll	Total first quarter payroll in \$1,000 with noise (2017)
13	APayroll	Total annual payroll in \$1,000 with noise (2017)
14	Establishments	Total number of establishments (2017)
15	Income	Per capita personal income in dollars (2018)
16	GDP	GDP in \$1,000 (2018)

17	Rural-urban_Continuum Code_2013	Integer scale from 1-9 describing a county's "rural"-ness (2013)
18	Urban_Influence_Code_2013	Integer scale from 1-9 describing a county's "urban"-ness (2013)
19	Economic_typology_2015	Integer scale from 0-5 describing a county's economic dependence (2015)
20	Less.than.a.high.school.diploma.2014.18	Number of adults who do not have a high school diploma (2018)
21	High.school.diploma.only.2014.18	Number of adults who just have a high school diploma (2018)
22	Some.college.or.associate.s.degree.2014.18	Number of adults with a college degree below bachelor's degree (2018)
23	Bachelor.s.degree.or.higher.2014.18	Number of adults who have at least a bachelor's degree (2018)
24	Percent.with.less.than.a.high.school.diploma.18	% adults who do not have a high school diploma (2018)
25	Percent.with.a.high.school.diploma.only.18	% adults who just have a high school diploma (2018)
26	Percent.completing.college.or..degree.18	% adults with a college degree below bachelor's (2018)
27	Percent.bachelor.s.degree.or.higher.18	% adults who have at least a bachelor's degree (2018)
28	POVALL_2018	Estimate of people of all ages in poverty (2018)
29	MEDHHINC_2018	Estimate of median household income (2018)
30	Density.Population	Land area density for population (2010)
31	Density.Housing.units	Land area density for population (2010)
32	Total_Male	Total male population
33	Total_Female	Total female population
34	Total_age18to64	Total population ages 18-64
35	Male_age18to64	Total male population ages 18-64
36	Female_age18to64	Total female population ages 18-64
37	Total_age65plus	Total population ages 65+
38	Male_age65plus	Total male population ages 65+
39	Female_age65plus	Total female population ages 65+
40	Total_age85plus	Total population ages 85+
41	Male_age85plus	Total male population ages 85+
42	Female_age85plus	Total female population ages 85+
43	Active Physicians	Active Physicians per 100000 Population 2018
44	Total Active Patient Care Physicians	Total Active Patient Care Physicians per 100000 Population (2018)
45	Active Primary Care	Active Primary Care Physicians per 100000 Population (2018)
46	Active General Surgeons	Active General Surgeons per 100000 Population (2018)
47	Active Patient Care General Surgeons	Active Patient Care General Surgeons per 100000 Population (2018)
48	Fraction Active Physician Female	Fraction of Active Physicians Who Are Female (2018)
49	Fraction IMGs	Fraction of Active Physicians Who Are International Medical Graduates (2018)
50	Fraction Physicians Age 60+	Fraction of Active Physicians 60+ (2018)
51	Anatomic/Clinical Pathology	Number of Anatomic/Clinical Pathologists
52	Anesthesiology	Number of Anesthesiologists
53	Cardiovascular Disease	Number of Cardiovascular Disease Specialists
54	Dermatology	Number of Dermatologists
55	Emergency Medicine	Number of Emergency Medicine Specialists
56	Family Medicine/General Practice	Number of Family Medicine/General Practitioners
57	Gastroenterology	Number of Gastroenterologists
58	General Surgery	Number of General Surgeons
59	Hematology/Oncology	Number of Hematologists/Oncologists
60	Internal Medicine	Number of Internal Medicine Specialists
61	Neurology	Number of Neurologists
62	Orthopedic Surgery	Number of Orthopedic Surgeons

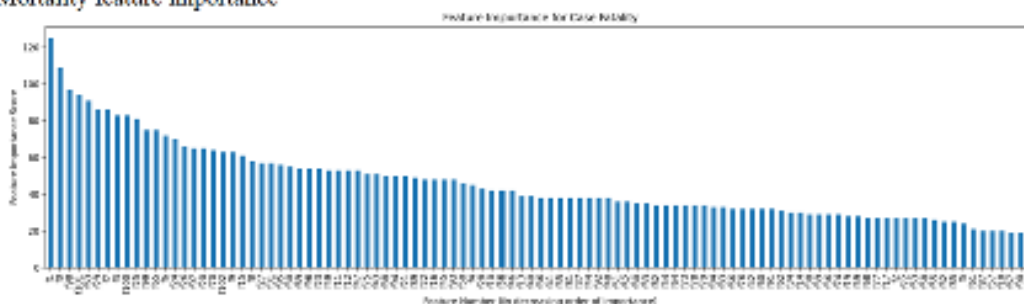
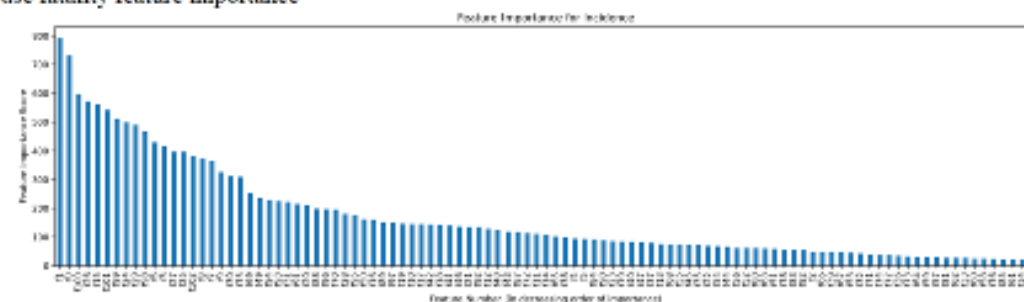
63	Otolaryngology	Number of Otolaryngologists
64	Pediatrics	Number of Pediatric Specialists
65	Plastic Surgery	Number of Plastic Surgeons
66	Psychiatry	Number of Psychiatrists
67	Radiology/Diagnostic Radiology	Number of Radiologists
68	Total nurse practitioners	Total number of nurse practitioners (2019)
69	Total physician assistants	Total number of physician assistants (2019)
70	Total Hospitals	Total number of hospitals (2019)
71	Internal Medicine Primary Care	Total number of Internal Medicine Primary Care specialists (2019)
72	Family Medicine/General Practice	Total number of Family Medicine/General Practice (2019)
73	Pediatrics Primary Care	Total number of Pediatrics Primary Care specialists (2019)
74	Obstetrics/Gynecology Primary Care	Total number of Obstetrics/Gynecology Primary Care specialists (2019)
75	Geriatrics Primary Care	Total number of Geriatric Primary Care specialists (2019)
76	Total Primary Care	Total number of Primary Care specialists (2019)
77	Psychiatry	Total number of Psychiatry specialists (2019)
78	Surgery	Total number of Surgery specialists (2019)
79	Anesthesiology	Total number of Anesthesiology specialists (2019)
80	Emergency Medicine	Total number of Emergency Medicine specialists (2019)
81	Radiology	Total number of Radiology specialists (2019)
82	Cardiology	Total number of Cardiology specialists (2019)
83	Oncology	Total number of Oncology (Cancer) specialists (2019)
84	Endocrinology	Total number of Endocrinology specialists (2019)
85	All Other Specialties	Total number of all other specialty specialists (2019)
86	Total Specialist Physicians	Total number of specialist physicians (2019)
87	ICU Beds	Total number of ICU Beds
88	UCR excluding arson	Total number of Uniform Crime Report crimes excluding arson
89	UCR including arson	Total number of Uniform crime report crimes including arson
90	Murder	Murder Frequency
91	Rape	Rape Frequency
92	Robbery	Robbery Frequency
93	Number of Aggravated Assaults	Number of Aggravated Assaults
94	Burglary	Burglary Frequency
95	Larceny	Larceny Frequency
96	Motor Vehicle Thefts	Motor Vehicle Theft Frequency
97	Arson	Arson Frequency
98	percent_smokers	% smokers
99	percent_physically_inactive	% physically inactive
100	percent_drinking	% drinking
101	percent_obese	% obese
102	percent_vaccinated	% vaccinated

Table A2. Classifiers and associated parameters. Default values were used for classifier parameters not listed above.

Classifier	Parameters
<i>k</i> -nearest neighbor	number of neighbors, weight
Linear discriminant analysis	prior
Quadratic discriminant analysis	prior
Decision tree ^a	max depth
Random forest ^a	max depth
Neural network ^a	number of hidden layers, activation function, gradient solver
XGBoost	max depth, min child weight, number of estimators

^a Stochastic classifiers have their random states set to 2019 for consistency.

APPENDIX B

(a) Incidence feature importance**(b) Mortality feature importance****(c) Case fatality feature importance****Figure B1.** Feature importance of incidence, mortality and case fatality as determined by the `xgboost.get_fscore()` function. Feature numbers correspond to features listed in Appendix A1.

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